Calculation Sheet - VBF-T6R1NOCEILING-CALC-01

Sheet chain: Inputs -> Capacity & Energy -> Energy Density -> N-P & Mass -> Process Params. All conclusion-grade values from tool outputs / parameter set.

VBF-T6R1NOCEILING-CALC-01

Sheet: Inputs

Quantity	Expression	Value	Unit	Source
Electrode strip height	parameter set	65.0	mm	Chen2020 Electrode height [mm]
Electrode strip width	parameter set	1580.0	mm	Chen2020 Electrode width [mm]
Electrode area	height x width	0.1027	m2	calc-energy output area_m2
Positive thickness	parameter set	75.6	um	params_r2_bp.json
Negative thickness	parameter set	85.2	um	params_r2_bp.json
Separator thickness	parameter set	9.0	um	params_r2_bp.json
Al collector thickness	parameter set	10.0	um	params_r2_bp.json
Cu collector thickness	parameter set	8.0	um	params_r2_bp.json
Stack thickness	sum of 5 layers	187.8	um	computed
Positive porosity	parameter set	0.3	-	params_r2_bp.json
Negative porosity	parameter set	0.2	-	params_r2_bp.json
Positive active volume fraction	parameter set	0.7	-	params_r2_bp.json
Negative active volume fraction	parameter set	0.8	-	params_r2_bp.json
Positive density (NMC811)	parameter set	3262.0	kg/m3	Chen2020
Negative density (graphite)	parameter set	1657.0	kg/m3	Chen2020
Positive particle radius	parameter set	5.22	um	Chen2020
Negative particle radius	parameter set	4.0	um	params_r2_bp.json
Cooling coefficient h	design lever (10-25)	10.0	W/m2 K	contract default
Nominal capacity (design)	loading-reconciled	5.6	Ah	params_r2_bp.json

Sheet: Capacity & Energy

Quantity	Expression	Value	Unit	Source
1C discharge capacity	run-pyamm 1C_discharge (SPMe)	5.322205945840946	Ah	cell/r2_bp_1c.json:capacity_ah
Discharge current (1C)	nominal capacity / 1h	5.6	A	protocol definition
Discharge energy	integral V x I dt over 1C discharge	18.601222347864425	Wh	cell/r2_bp_energy.json:energy_wh
Voltage window	parameter set cut-offs	2.5 - 4.2	V	Chen2020 Lower/Upper voltage cut-off
Discharge-time midpoint voltage	V at discharge-time midpoint	3.921427550822483	V	cell/r2_bp_energy.json:midpoint_voltage_v
DC resistance	10%-discharge definition	0.0003636012830776636	ohm	cell/r2_bp_energy.json:dcr_ohm
Power density	DCR-based (calc-energy definition)	283313.20463461475	W/kg	cell/r2_bp_energy.json:power_density_w_kg

Sheet: Energy Density

Quantity	Expression	Value	Unit	Source
Stack volume	stack thickness x area	1.9287060000000004e-05	m3	calc-energy output volume_m3 (= 19.29 cm3)
Volumetric energy density	energy / volume (contract: electrolyte excluded)	964.4405289279143	Wh/L	cell/r2_bp_energy.json:energy_density_wh_l
Cell mass (contract)	electrodes + CCs + separator	0.03965656259500001	kg	calc-energy output mass_kg (39.66 g)
Gravimetric energy density	energy / mass	469.0578590442357	Wh/kg	cell/r2_bp_energy.json:energy_density_wh_kg
Criterion ED	energy_density_wh_l >= 950	PASS (964.44)	-	log entry 0 + log-evaluate

Sheet: N-P & Mass

Quantity	Expression	Value	Unit	Source
Positive areal discharge capacity	c_max x active frac x thickness x delta_x x 26.8 Ah/mol	65.33326743551999	Ah/m2	computed from parameter set
Negative areal discharge capacity	c_max x active frac x thickness x delta_x x 26.8 Ah/mol	54.47128815360001	Ah/m2	computed from parameter set
N/P ratio	negative areal capacity / positive areal capacity	0.833745047381258	-	negative-limited design (Chen2020 characteristic kept)
Positive active mass	layer_kg_m2 x area	17.728591608	g	cell/r2_bp_energy.json:layer_kg_m2
Negative active mass	layer_kg_m2 x area	11.599053024	g	cell/r2_bp_energy.json:layer_kg_m2
Al collector mass	layer_kg_m2 x area	2.7729000000000004	g	cell/r2_bp_energy.json:layer_kg_m2
Cu collector mass	layer_kg_m2 x area	7.361535999999999	g	cell/r2_bp_energy.json:layer_kg_m2
Separator mass	layer_kg_m2 x area	0.19448196299999998	g	cell/r2_bp_energy.json:layer_kg_m2
Total mass (contract)	sum of the 5 layers	39.656562595000004	g	calc-energy output mass_kg x 1000

Sheet: Process Params

Quantity	Expression	Value	Unit	Source
Positive areal density	thickness x (1-porosity) x density	172.62503999999998	g/m2	cell/r2_bp_energy.json:layer_kg_m2
Negative areal density	thickness x (1-porosity) x density	112.94112	g/m2	cell/r2_bp_energy.json:layer_kg_m2
Positive compaction density	density x active fraction / 1000	2.2833999999999994	g/cm3	computed (3262 x 0.70 / 1000 = 2.28)
Negative compaction density	density x active fraction / 1000	1.3256000000000001	g/cm3	computed (1657 x 0.80 / 1000 = 1.33)
Electrolyte pore volume	sum(thickness x porosity) x area	4.513665	cm3	computed from parameter set
Electrolyte fill amount	pore volume x electrolyte density (1.2 g/cm3 literature)	5.416397999999999	g	annotated literature value
Formation recommendation	-	0.1C CC to 4.2 V, 25 C, 2 cycles	-	design recommended value; production tuning required
Binder / conductive additive	-	see BOM (literature defaults, annotated)	-	parameter set has no inert-volume params; production must rebalance active fraction